习题 8.2

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- (5) 提示

  $$\int \left(\frac{1}{\sqrt{3-x^2}} + \frac{1}{\sqrt{1-3x^2}} \right) dx = \int \frac{1}{\sqrt{3-x^2}} dx + \int \frac{1}{\sqrt{1-3x^2}} dx$$

  分别令

  $$x = \sqrt{3} sint$$
  $$x = \frac{1}{\sqrt{3}} sint$$

- (12)

  $$\begin{aligned} \int \frac{1}{1+sinx} dx &= \int \frac{1-sinx}{1-sin^2x} dx \\ &= \int \frac{1-sinx}{cos^2x} dx \\ &= \int \frac{1}{cos^2x} dx - \int \frac{sinx}{cos^2x} dx \\ &= \int \frac{1}{cos^2x} dx + \int \frac{1}{cos^2x} d(cosx) \\ &= tanx - cos^{-1}x + C \\ &= tanx - secx + C \end{aligned}$$

- (13)

$$\begin{aligned}\int cscxdx &= \int \frac{cscx(cotx-cscx)}{cotx-cscx}dx \\ &= \int \frac{d(cotx-cscx)}{cotx-cscx} \\ &= \ln|cotx-cscx|+C\end{aligned}$$

- (15)

$$\begin{aligned}\int \frac{x}{4+x^4}dx &= \frac{1}{4}\int \frac{x}{1+(\frac{x^2}{2})^2} \\ &= \frac{1}{4}\int \frac{1}{1+(\frac{x^2}{2})^2}d(\frac{x^2}{2}) \\ &= \frac{1}{4}tan\frac{x^2}{2}+C\end{aligned}$$

- (19)

$$\begin{aligned}\int \frac{dx}{x(1+x)} &= \int \frac{dx}{x} - \int \frac{dx}{1+x} \\ &= \ln|x| - \ln|1+x| + C \\ &= \ln\left|\frac{x}{1+x}\right| + C\end{aligned}$$

- (21)

  提示：

$$\int cos^5xdx = \int (1-sin^2x)^2d(sinx)$$

- (22)

$$\begin{aligned}\int \frac{dx}{sinxcosx} &= \int \frac{dx}{tanxcos^2x} \\ &= \int \frac{d(tanx)}{tanx} \\ &= \ln|tanx|+C\end{aligned}$$

- (23)

$$\begin{aligned}\int \frac{dx}{e^x+e^{-x}} &= \int \frac{dx}{\frac{(e^x)^2+1}{e^x}} \\ &= \int \frac{e^x}{(e^x)^2+1}dx \\ &= \int \frac{1}{(e^x)^2+1}d(e^x) \\ &= arctan\ e^x + C\end{aligned}$$

- (25)

$$\begin{aligned}\int \frac{x^2+2}{(x+1)^3}dx &= \int \frac{(x+1)^2-2(x+1)+3}{(x+1)^3}dx \\ &= \int \frac{(x+1)^2-2(x+1)+3}{(x+1)^3}d(x+1) \\ &= \int \frac{1}{x+1}d(x+1) - 2\int \frac{1}{(x+1)^2}d(x+1) + 3\int \frac{1}{(x+1)^3}d(x+1) \\ &= \ln|x+1| + 2(x+1)^{-1} - \frac{3}{2}(x+1)^{-2} + C\end{aligned}$$

- (26)

  提示：令 $x = atant$

- (29)

  令 $x = t^6(t \geq 0, t \neq 1), dx = 6t^5dt$。

$$\begin{aligned}\int \frac{\sqrt{x}}{1-\sqrt[3]{x}}dx &= \int \frac{t^3}{1-t^2}6t^5dt \\ &= 6\int \frac{t^8}{1-t^2}dt \\ &= 6\int \frac{t^8-1+1}{1-t^2}dt\end{aligned}$$

  我们有

$$\begin{aligned}t^8-1 &= (t^2)^4-(1^2)^4 \\ &= (t^2-1)[(t^2)^3+(t^2)^2+t^2+1]\end{aligned}$$

所以

$$
\begin{aligned}
6\int\frac{t^8-1+1}{1-t^2}dt &= 6\int -t^6-t^4-t^2-1+\frac{1}{1-t^2}dt \\
&= 6\left(-\frac{1}{7}t^7-\frac{1}{5}t^5-\frac{1}{3}t^3-t+\frac{1}{2}\ln\left|\frac{t+1}{t-1}\right|\right)+C \\
&= -\frac{6}{7}t^7-\frac{6}{5}t^5-2t^3-6t+3\ln\left|\frac{t+1}{t-1}\right|+C \\
&= -\frac{6}{7}x^{\frac{7}{6}}-\frac{6}{5}x^{\frac{5}{6}}-2x^{\frac{1}{2}}-6x^{\frac{1}{6}}+3\left|\frac{x^{\frac{1}{6}}+1}{x^{\frac{1}{6}}-1}\right|+C
\end{aligned}
$$

- (30)

  令 $t=\sqrt{x+1}, x=t^2-1, dx=2tdt$。

$$
\begin{aligned}
\int\frac{\sqrt{x+1}-1}{\sqrt{x+1}+1}dx &= \int\frac{t-1}{t+1}2tdt \\
&= 2\int\frac{(t-1)t}{t+1}dt
\end{aligned}
$$

说明 1. 我们有 $t+a$，使得

$$
\begin{aligned}
(t+1)(t+a)+b &= (t-1)t \\
t^2+at+t+a+b &= t^2-t \\
t^2+(a+1)t+a+b &= t^2-t
\end{aligned}
$$

于是

$$
\begin{cases}
a+1=-1 \\
a+b=0
\end{cases}
$$

可得

$$a=-2, b=2$$

于是

$$(t+1)(t-2)+2=(t-1)t$$

所以

$$
\begin{aligned}
2\int \frac{(t-1)t}{t+1}dt &= 2\int \frac{(t+1)(t-2)+2}{t+1}dt \\
&= 2\int t-2+\frac{2}{t+1}dt \\
&= 2(\frac{1}{2}t^2-2t+2\ln|t+1|)+C \\
&= t^2-4t+4\ln|t+1|+C
\end{aligned}
$$

剩下的部分，略

- (31)

  令 $t=1-2x, x=\frac{1-t}{2}, dx=-\frac{1}{2}dt$。

$$
\begin{aligned}
\int x(1-2x)^{99}dx &= -\frac{1}{2}\int \frac{1-t}{2}t^{99}dt \\
&= -\frac{1}{4}\int (1-t)t^{99}dt \\
&= -\frac{1}{4}\int t^{99}-t^{100}dt \\
&= -\frac{1}{4}(\frac{1}{100}t^{100}-\frac{1}{101}t^{101})+C \\
&= -\frac{1}{400}(1-2x)^{100}+\frac{1}{404}(1-2x)^{101}+C
\end{aligned}
$$

- (32)

  提示：

$$
\frac{1}{x(1+x^n)} = \frac{1}{x}-\frac{x^{n-1}}{1+x^n}
$$

- (33)

$$
\begin{aligned}
\int \frac{x^{2n-1}}{x^n+1}dx &= \frac{1}{n}\int \frac{x^n}{x^n+1}d(x^n) \\
&= \frac{1}{n}\int 1-\frac{1}{x^n+1}d(x^n) \\
&= \frac{1}{n}(x^n-\ln|x^n+1|)+C
\end{aligned}
$$

- (35)

$$\begin{aligned}\int \frac{\ln 2x}{x\ln 4x}dx &= \int \frac{\ln 4x - \ln 2}{x\ln 4x}dx \\ &= \int \frac{1}{x} - \frac{\ln 2}{x\ln 4x}dx \\ &= \int \frac{1}{x}dx - \ln 2\int \frac{1}{x\ln 4x}dx \\ &= \int \frac{1}{x}dx - \ln 2\int \frac{1}{\ln 4x}d(\ln 4x) \\ &= \ln x - \ln 2 \cdot \ln|\ln 4x| + C\end{aligned}$$

- (36)

令 $x = sect, dx = sect \cdot tantdt$。【这里的 $t \in (0, \frac{\pi}{2}) \cup (\pi, \frac{3\pi}{2})$，保证 $\sqrt{tan^2t}$ 的绝对值可以去掉。考试时可以不写这部分】

$$\begin{aligned}\int \frac{dx}{x^4\sqrt{x^2-1}} &= \int \frac{sect \cdot tant}{sec^4x \cdot tant}dt \\ &= \int \frac{1}{sec^3x}dt \\ &= \int cos^3tdt \\ &= int(1-sin^2t)d(sint) \\ &= sint - \frac{1}{3}sin^3t + C\end{aligned}$$

然后通过三角图形，把 t 换成 x 即可。